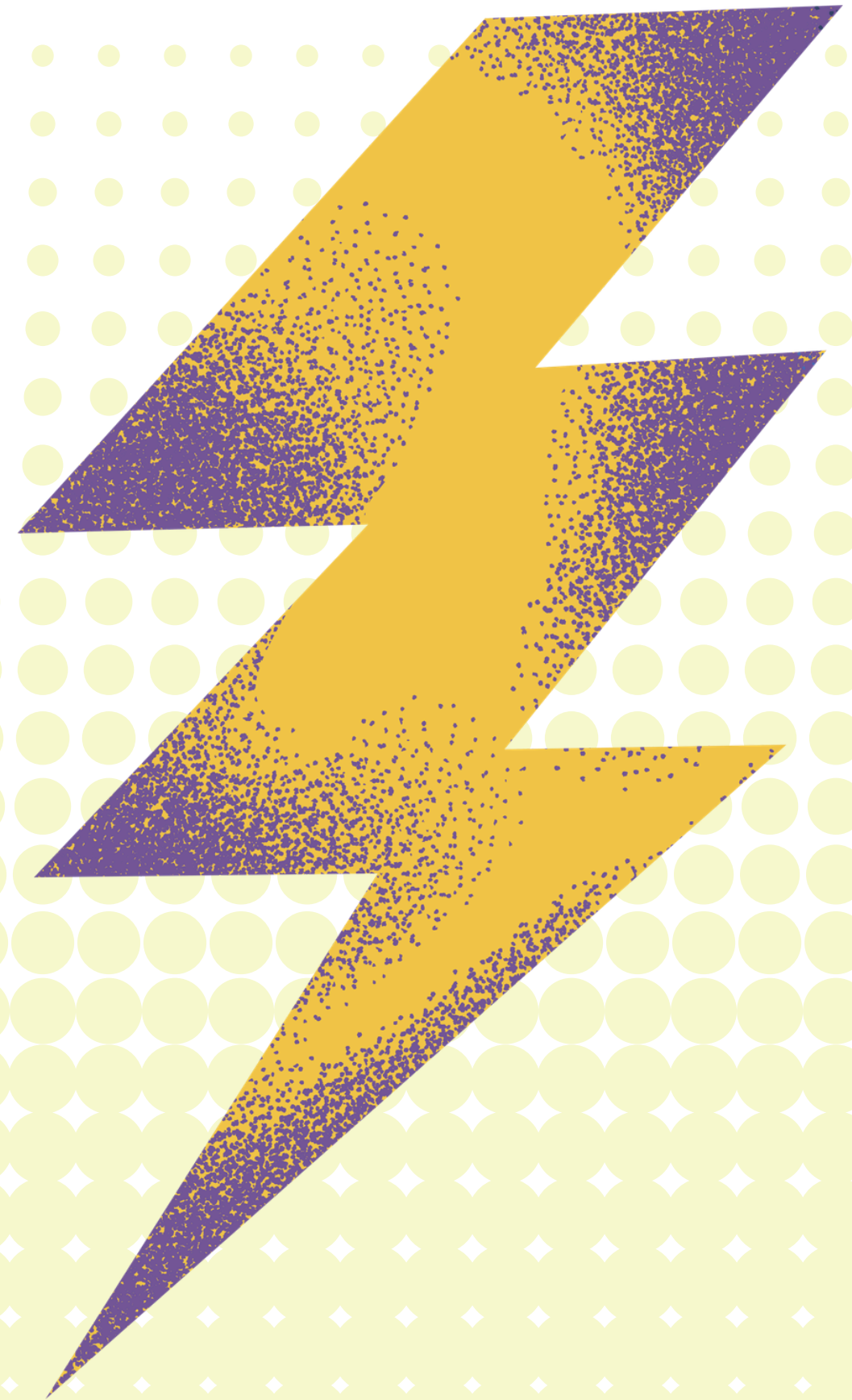


# STATIC ELECTRICITY



# WHAT IS STATIC ELECTRICITY?

Static electricity is the form of electricity resulting from the imbalance between positive and negative charges within a material that occurs when electrons (the negatively charged particles in an atom) move from one material to another.





# BASIC CONCEPTS

When different atoms make contact with one another, electrons can transfer between them.

The material shedding electrons also loses negative charge, and it becomes positively charged when there are more protons than electrons. So, a material that gains electrons becomes negatively charged. As more electrons move from one material to the other—due to repeated contact between them—additional negative charge will build up in a process that is referred to as the triboelectric effect.



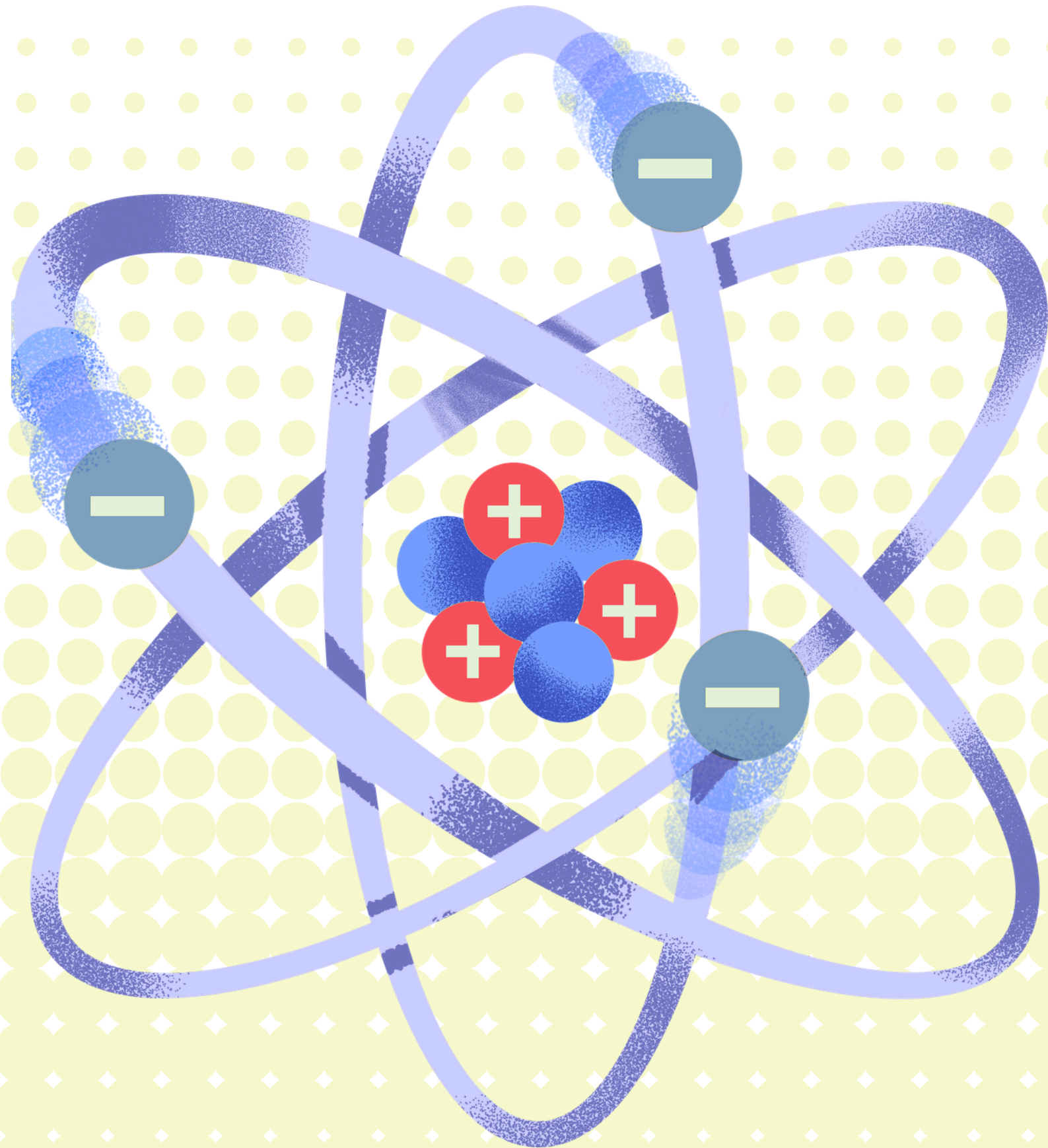
# EXAMPLES

1

When a person's feet move across a carpet, they acquire equal and opposite charges, and an attractive force develops between them. The atoms in the person's body strip away electrons from the carpet and leave behind a positive charge on the carpet's fibres. The electrons become isolated in the atoms of the person's body.

2

A lightning strike happens when part of a cloud builds up excess electrical charge. Inside the cloud, small hail particles form as moisture freezes. As these particles grow, move, and collide, they transfer charge, causing positive and negative charges to separate. Air acts as an insulator between these charged areas and between the cloud and the ground. When the charge becomes strong enough to overcome the air's insulating effect, electricity is released as a lightning discharge between clouds or between the cloud and the ground.



# COULOMB'S LAW

$$F_e = k_e \frac{|q_1| |q_2|}{r^2}$$

where  $F_e$  is the electric force

$k_e$  is the Coulomb constant

$$k_e = 8.9875 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$$

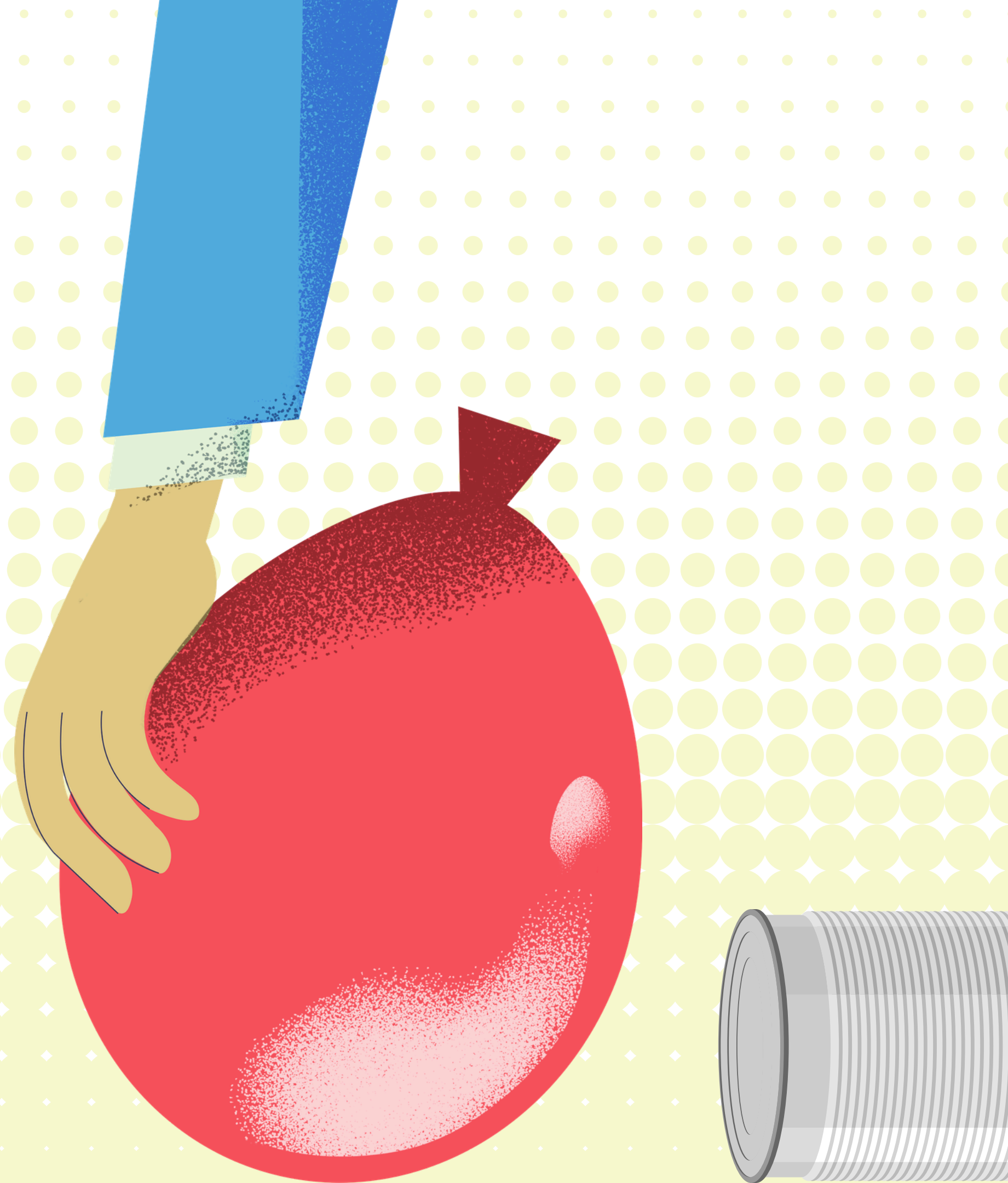
$q_i$  is the charge on object  $i$

and  $r$  is the distance between the charges

# QUESTION 1

Two balloons are rubbed on wool and become charged. One balloon has a charge of  $+4.5\ \mu\text{C}$ , and the other has  $+2.5\ \mu\text{C}$ . They are hanging  $0.80\ \text{m}$  apart in the air.

How much electric force do the balloons exert on each other?



# ANSWER 1

$$F = (8.99 \times 10^9) \frac{(4.5 \times 10^{-6})(2.5 \times 10^{-6})}{(0.80)^2}$$

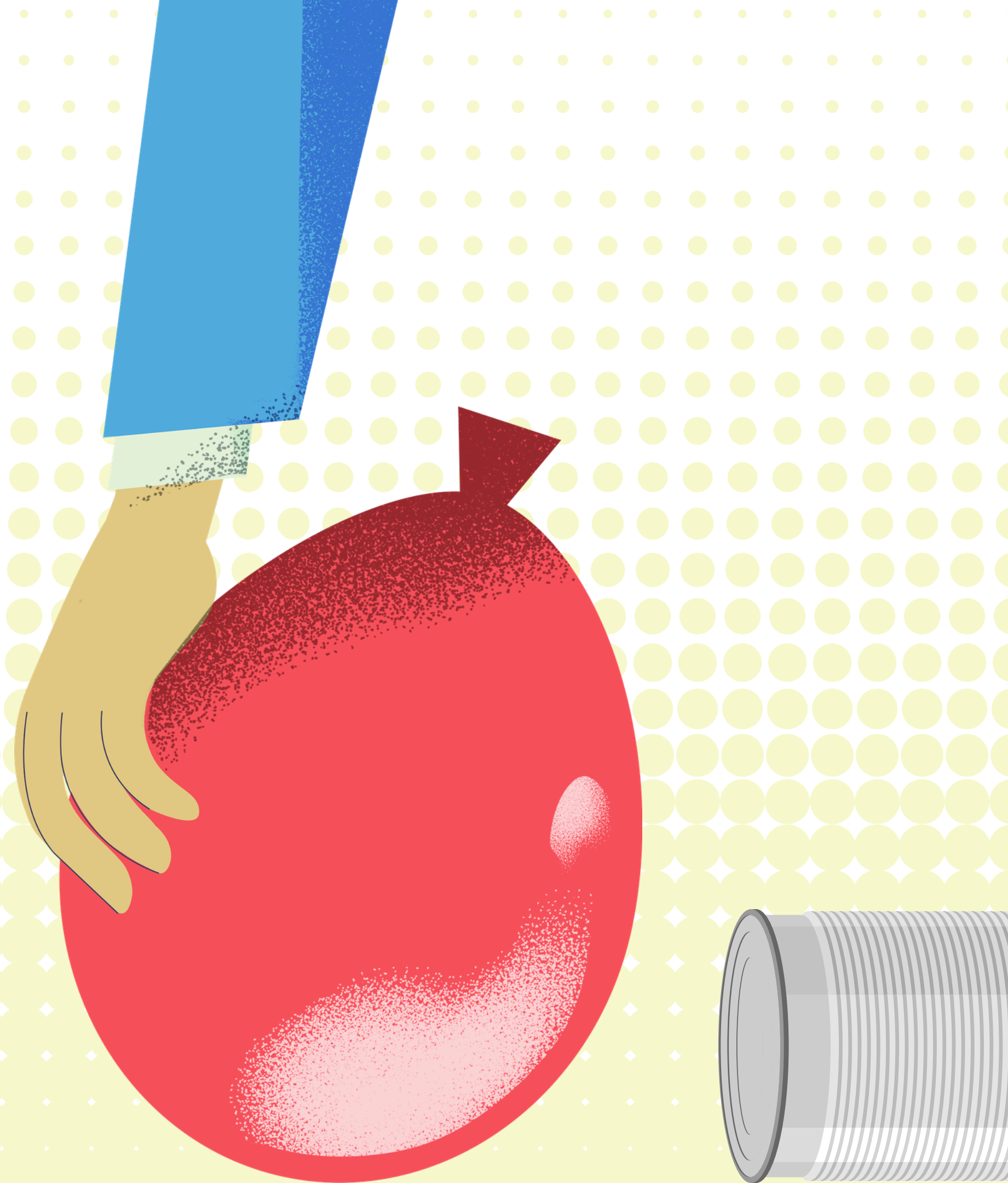
$$(4.5 \times 10^{-6})(2.5 \times 10^{-6}) = 11.25 \times 10^{-12}$$

$$8.99 \times 10^9 \times 11.25 \times 10^{-12} = 0.101 \text{ N}$$

$$0.80^2 = 0.64$$

$$F = \frac{0.101}{0.64}$$

$$F = 0.158 \text{ N} \approx 0.16 \text{ N (repulsive)}$$



# ANSWER 2

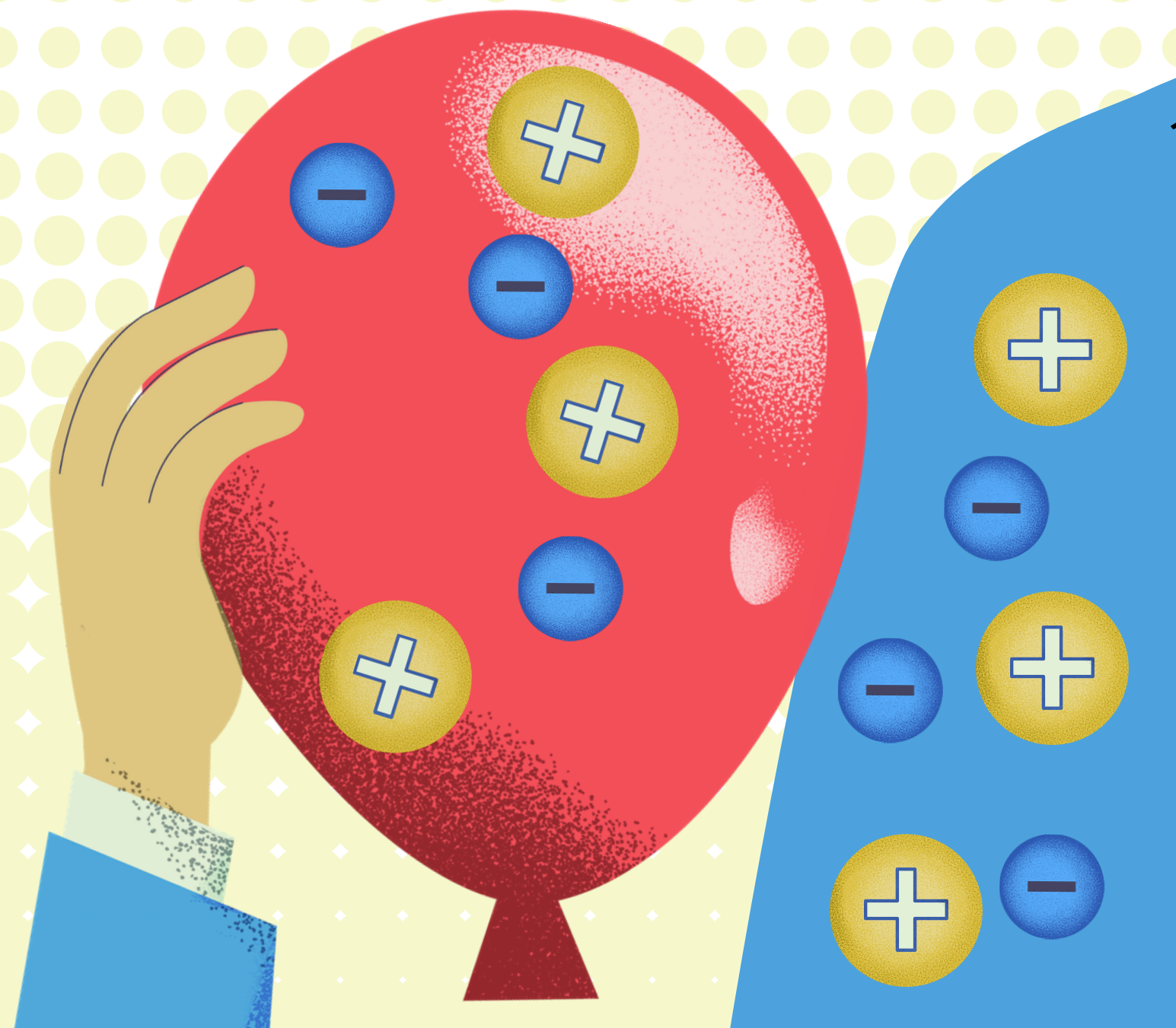
$$F = (8.99 \times 10^9) \frac{(8.0 \times 10^{-6})(4.0 \times 10^{-6})}{(0.40)^2}$$

$$(8.0 \times 10^{-6})(4.0 \times 10^{-6}) = 32 \times 10^{-12}$$

$$8.99 \times 10^9 \times 32 \times 10^{-12} = 0.288 \text{ N}$$

$$0.40^2 = 0.16$$

$$F = \frac{0.288}{0.16} = 1.8 \text{ N (attractive)}$$



THANK  
YOU

